



Disaster Management

Flooding in Ghana

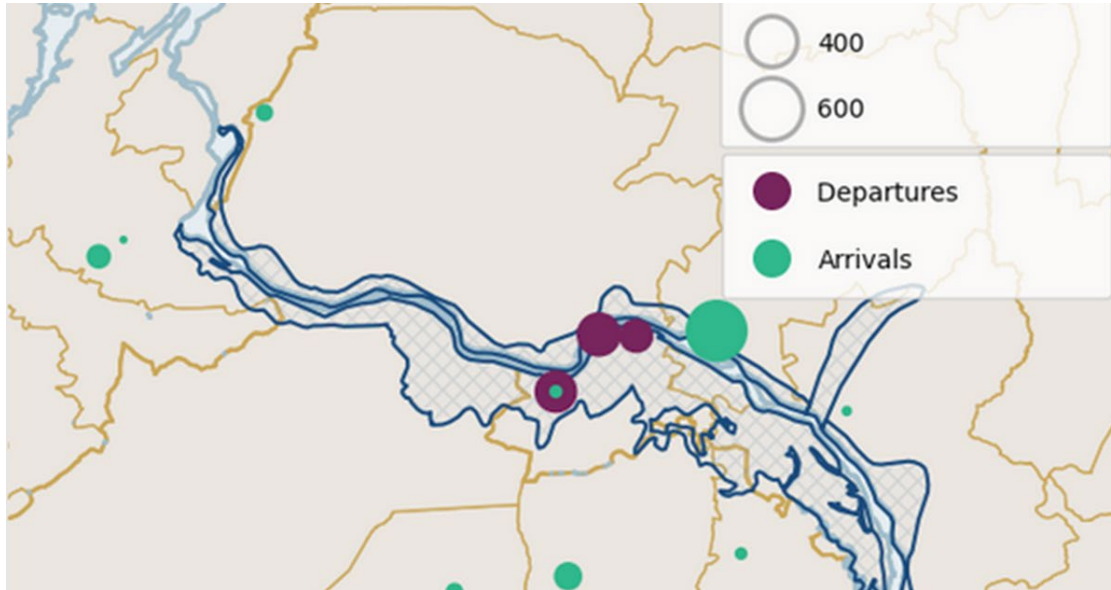
Course 2- Module 2- Use case
Disaster Management

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NOMMON



Flooding from the Akosombo Dam, October 2023



Objectives

- Supporting the National Disaster Management Organisation (NADMO) in Ghana to understand where people have moved to after the flooding caused by the release of excess water from behind the Akosombo Dam on 11 October 2023

Flooding from the Akosombo Dam, October 2023

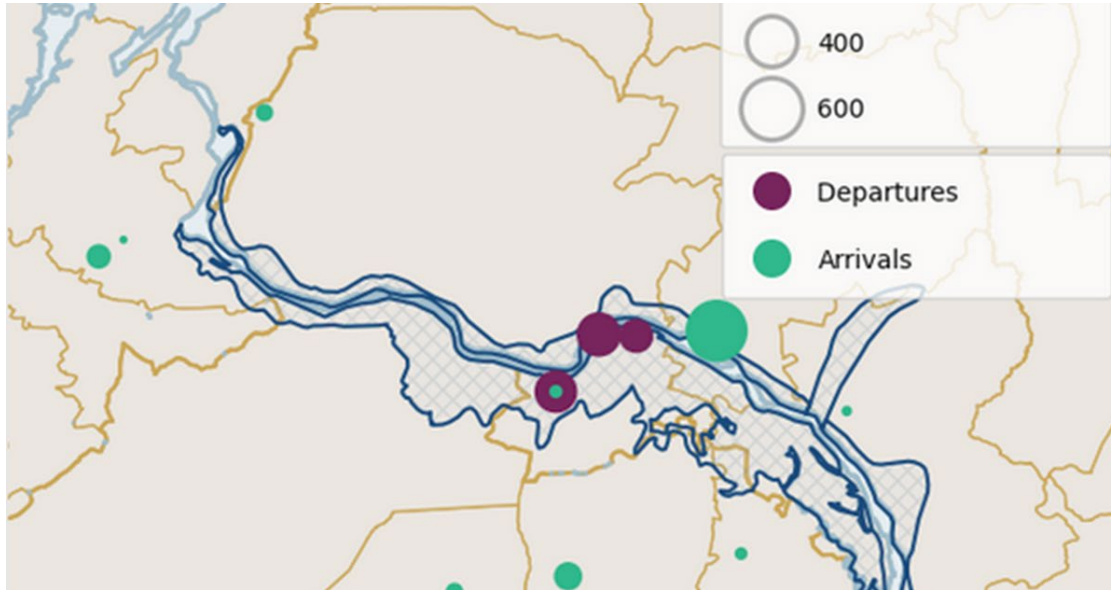


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**GHANA
STATISTICAL
SERVICE**

Flooding from the Akosombo Dam, October 2023



Methodology

Home locations:

- For crisis response computing changes in location quickly is vital for informing actors on the ground
- We use the modal last call of the day over the last week as the home location of a subscriber for the purpose of crisis reporting

Methodology

Home locations:

day	<i>d-6</i>	<i>d-5</i>	<i>d-4</i>	<i>d-3</i>	<i>d-2</i>	<i>d-1</i>	<i>d</i>
last call	A	A	B	B	B	A	A

This subscriber has been assigned location A as their home location as they have made 4 last calls over the last week at that location.

Methodology

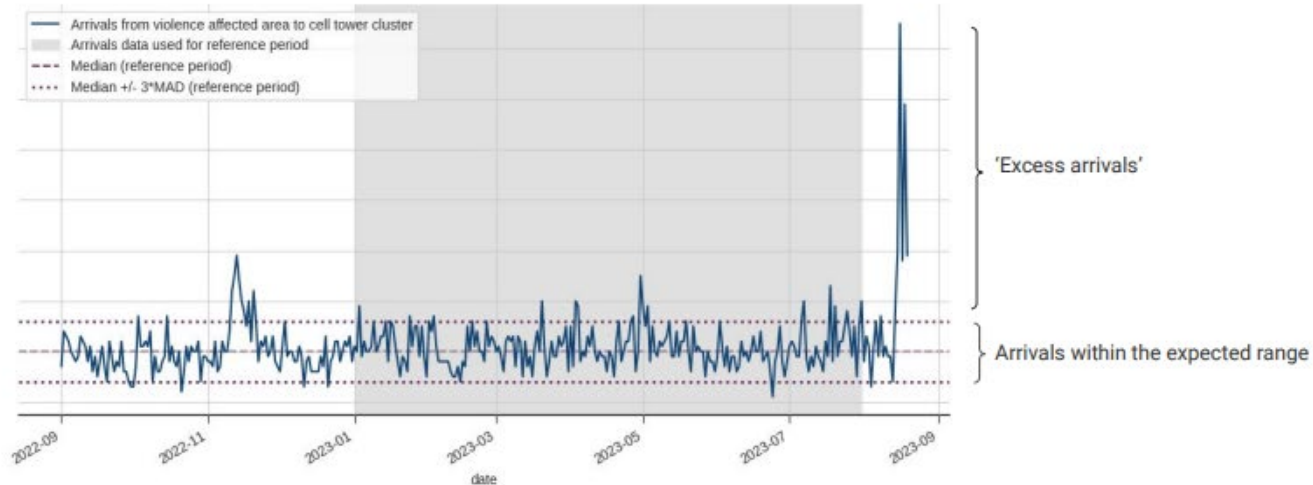
Relocations:

Home relocations from Location A to Location B are computed as the total number of subscribers whose home location changes from A to B. These are aggregated per (origin, destination) pair.

Methodology

Methodology: Excess Arrivals

Graph 2: Computation of excess arrivals. Dotted lines show the 'normal range' of arrivals, anything outside this range is considered as 'above normal'.



Results

Report #1

Most recent data: 22 October 2023
Published: 07 November 2023

Ghana | October 2023 Lower Volta River Floods

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Long- and medium-distance displaced subscribers from flood-affected areas of the Lower Volta River: Situation as of 22 October 2023

Executive summary

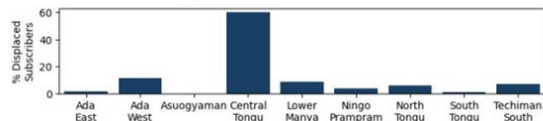
The release of excess water from behind Akosombo Dam on 11 October 2023, following heavy and unpredictable rainfall, has resulted in the flooding on communities in nine districts along the Lower Volta River. Large displacements have been reported as a result of the flooding (over 26,000, source: [ELERCC](#)). In this report, we focus on medium- (5–25km) and long-distance displacements (greater than 25km). We estimate that 1,430 Vodafone Ghana subscribers have been displaced from the areas around cell clusters in the affected districts. This represents only a portion of the total number of displaced persons, as these figures do not include those without a subscription to Vodafone Ghana or those who were displaced very short distances (less than 2km). We observe very few long distance displacements or displacements to urban centres, such as the Accra Metro area. The largest displacements observed were from the area around Mepe and Aveyime in North Tongu, to Mafi-Adidome in Central Tongu. We estimate that approximately 853 additional subscribers are residing in the area around Mafi-Adidome. We also observe that the increase in the number of subscribers displaced from the affected area has slowed. However, we do not yet observe any indications that subscribers are returning to their previous home locations.

Where have residents of the affected area been displaced to around Ghana?

Origins and destinations of displaced subscribers

The bubbles on Map 1 (right) correspond to the **number of displaced subscribers** who have departed areas affected by flooding (purple) and the areas which have received displaced subscribers (green). The density of cell clusters means that we do not observe very short-distance displacements (less than 2km). **The largest number of displaced subscribers depart from the areas around Mepe and Aveyime in North Tongu. The cluster with the largest number of arriving displaced subscribers is close to Mafi-Adidome in Central Tongu.** We observed very few displaced subscribers outside of Eastern, Volta or Great Accra regions.

Graph 1 (below) shows that the **majority of displaced subscribers (60.1%) relocated from the affected area to Central Tongu.**



Graph 1: Percentage of the total displaced subscribers from the affected area who have relocated to each district



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The mobile operator data provided to Flowminder for this report was fully anonymised. No personal data was made available at any time (e.g. an individual's identity, demographics, location, contacts or movements). All results are aggregated (for example, subscriber density in a given municipality), meaning that they do not contain any information about individual subscribers.

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Results

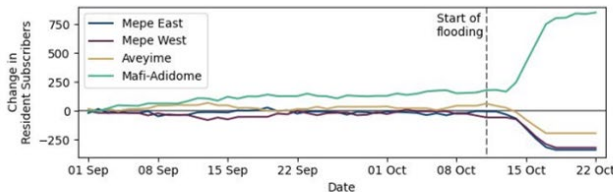
Flows of displaced subscribers

The arcs on Map 2 (right) correspond to the **number of displaced subscribers** who relocated from one area to another (purple indicates the origin of the flow and green indicates the destination). **Most observed displacements are to areas in neighbouring or nearby districts.** We observe **very few long distance displacements** or displacements to urban centres such as the Accra Metro area. **The largest flows of displaced subscribers are from the area around Mepe and Aveyime in North Tongu, to Mafi-Adidome in Central Tongu.** Subscribers residing near Mepe and Aveyime also relocated to the Great Accra region.

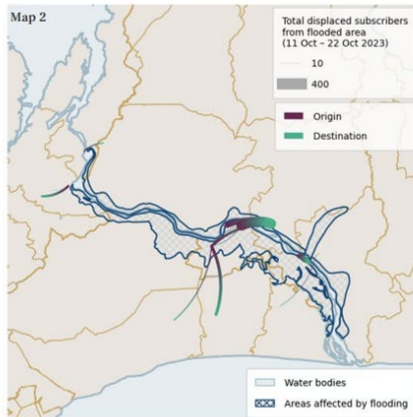
How have the numbers of subscribers residing in the affected area changed?

We observe **large changes in the number of residents in North and Central Tongu**, in the Volta region (Graph 2). **The number of subscribers residing in the vicinity of Mafi-Adidome increased rapidly in the period following the start of the flooding**, but this increase has since slowed. As of 22 October 2023, **an additional 853 subscribers are residing in the area of Mafi-Adidome.** We can observe contemporaneous **decreases in the number of residents in the area around Mepe and Aveyime in North Tongu.**

We do not yet observe the population in these areas starting to return to their pre-flood baseline, which indicates the **subscribers remain displaced** and are not yet returning to their previous home locations.



Graph 2: Percentage change in the number of resident subscribers in area around Mepe and Aveyime-Battor in North Tongu and Mafi-Adidome in Central Tongu.



Conclusions

Based on anonymous aggregated mobile phone data from Vodafone Ghana, **most displacements caused by the flooding on the Lower Volta River may have been local** (less than 25km). Furthermore, most displaced subscribers are likely to remain within the affected or neighbouring districts.

The number of new displacements may also have slowed since the initial large displacement. However, we do not yet observe the population around the cell clusters starting to return to their pre-flood baseline, indicating that **subscribers remain displaced.**

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Complete Case Study

Module 2 Use Case - Disaster Management in Ghana

Background

Origins

- Discussions between Vodafone Ghana (now Telecel Ghana) and Ghana Ministry of Health first triggered by the Ebola crisis in 2016, but a partnership didn't materialise at the time.
- A meeting between Ghana Statistical Services and Flowminder in 2017 led to new discussion and the further development of previous discussions with Vodafone Ghana.
- Data for Good Partnership between Vodafone Ghana and GSS, supported by Flowminder, launched in 2018.

Stakeholders

The three main partners involved in the initiative who have signed a partnership agreement to work together are:

1. Telecel Ghana (previously, Vodafone Ghana)
2. Ghana Statistical Service (GSS)
3. Flowminder Foundation

The country's two regulatory bodies, the National Communications Authority and the Data Protection Commission were also involved in granting permission for the data to be processed.

Other bodies, including the trade body for Mobile Network Operators (Ghana Chamber of Telecommunications) and the Global Partnership for Sustainable Development Data, were involved in the initial discussions prior to the establishment of the partnership which became known as the Data for Good Partnership.



The GDF-MPD window is a multi-partner window funded by the Spanish Government, Ministry of Economy, Commerce and Business. The GDF-MPD program is jointly managed by DEC (DECDG, DIME), Poverty and Digital Development Global Practices in technical partnership with the International Telecommunications Union (ITU).



Mobile Phone Data for Policy Programme (WB -GDF)
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